Coping with the parallax effect in 2D X-ray detectors

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Abstract

The parallax effect in X-ray area detectors arises when X-rays emitted from a point source (e.g. a scattering sample) traverse a thick sensitive layer, causing a shift in the apparent position of the detected events. This shift displaces and broadens the diffraction peaks, which degrades the angular resolution and introduces artefacts, especially in powder diffraction and pair distribution function (PDF) measurements. This becomes especially apparent in thick sensors, where X-rays with incidence angles above 30° may cross several pixels, leading to discrepancies between observed and expected scattering angles (2θ) for a given peak. In this contribution, the parallax effect is simulated by ray tracing based on the Beer–Lambert law and validated on synthetic data. Several correction strategies are compared, in particular a new integral method, implemented in the pyFAI library, which corrects the pixel positions regardless of the detector position and orientation. The corrections are assessed against experimental data collected

from reference samples (silicon, ceria) on the D2AM beamline (ESRF-BM02) using Eiger2 detectors equipped with either a 450 μm silicon sensor or a 750 μm cadmium telluride sensor.

1. Introduction

Experimental setups are usually calibrated at synchrotron sources using the diffraction pattern of a reference material, a properly characterized powder sample (provided by NIST, such as LaB_6 or CeO_2). Those samples show well-defined concentric Debye–Scherrer rings in their diffraction images that are analyzed with tools such as pyFAI-calib2 (Kieffer *et al.*, 2020), Dawn (Filik *et al.*, 2017) or GSAS-II (Toby & Von Dreele, 2013). These tools assign the observed rings to the d-spacings of the families of Miller planes and refine the geometry model, which is based on Bragg's law:

$$2d\sin(\theta) = n\lambda \tag{1}$$

Best calibrations are obtained when using as many rings and as many control points per ring as possible. Nevertheless, the quality of this fit has proven to degrade when including rings with $2\theta > 30°$ into the calibration. This degradation is associated with volumetric absorption of X-ray photons in the sensitive layer of the detector, simply referred to as the parallax effect in the rest of the document. The influence of this effect on the point-spread function of thick silicon sensors has recently been quantified by Monte Carlo simulation in the context of coherent diffraction imaging (Kuster *et al.*, 2025).

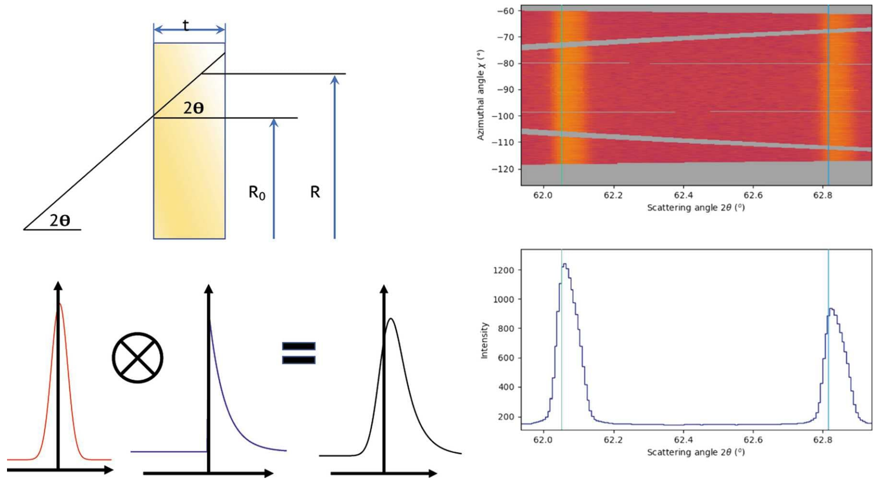

Fig. 1. Top left: detector in orthogonal geometry. Bottom left: illustration of convolution equation and the resulting line shape. Top right: azimuth–2θ map for LaB_6 pattern measured at a sample-to-detector distance of 138 mm at 19.5 keV at SNBL beamline (ESRF). Bottom right: high-angle diffraction lines together with theoretical positions. There is no shift of the line position and no asymmetry for the lines at low angles. Vertical lines show the expected positions of the diffraction peaks. The data were processed by pyFAI-calib2. Figure reused with permission from Chernyshov *et al.* (2021).

The parallax effect has four main consequences on the obtained powder diffraction pattern, illustrated in figure 1:

- Peak position shift, usually towards larger scattering angles.
- Peak broadening, usually increasing with the incidence angle.
- Peak asymmetry (the maximum lies at lower incidence angles).
- Peak profile distortions, which are likely to jeopardize Rietveld refinements.

1.1. Visualization of the parallax effect

Since the parallax effect depends on the detector sensor, it is possible to visualize the effect by recording the same diffraction frame with two detectors differing only by their sensor material. The diffraction patterns of silicon (SRM 640f) (Black *et al.*, 2020) and CeO_2 (SRM 674b) (National Institute of Standards and Technology, 2012) powders were collected on the D2AM beamline (ESRF-BM02) with a monochromatic beam at 25 keV and are compared in figure 2. Eiger2 (Casanas *et al.*, 2016) 4 mega-pixel detectors were used, either with a silicon sensor (450 μm thickness, $\mu = 6.87$ cm^{-1}, detection efficiency $\eta = 27\%$) or a cadmium telluride sensor (750 μm thickness, $\mu = 75$ cm^{-1}, $\eta = 99\%$). The simple replacement of the sensor is of course not possible in such a detector, so two different detectors were used. In order to avoid any artefact originating from the calibration, the plot in figure 2 actually contains only the signal of the column of pixels where the direct beam hits the detector. To compensate for the difference in detection efficiency, the signal of the silicon sensor has been multiplied by 3.7 so that the intensities of the two detectors are comparable.

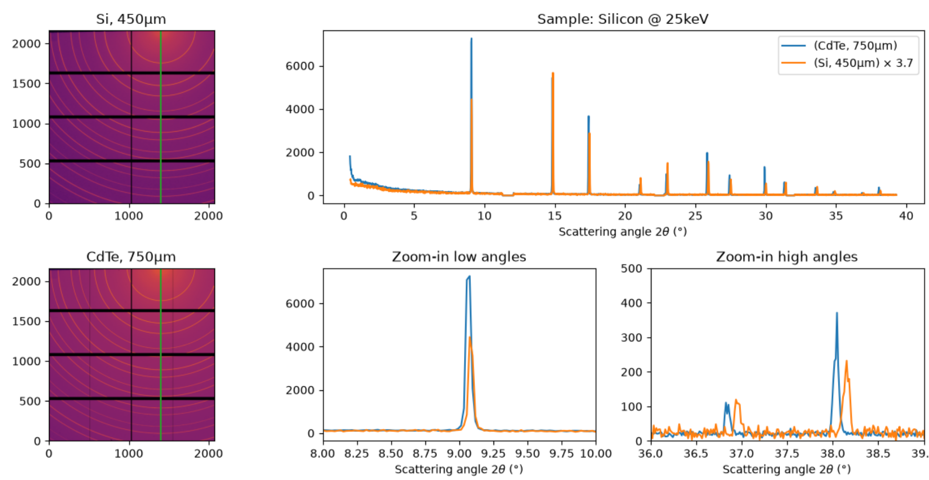

Fig. 2. Comparison of the signal recorded with the Si (450 μm) and CdTe (750 μm) sensors for the powder diffraction of silicon (SRM640) at 25 keV. Left: diffraction images, with the column of pixels passing through the direct beam marked by the green line. Right: intensity along this column as a function of the scattering angle 2θ, with zooms at low and high angles. The signal of the Si sensor is multiplied by 3.7 to make the intensities comparable. The diffraction peaks at low angles superimpose, whereas the peaks recorded at high angles are displaced.

In this contribution we first describe the various attempts to address the parallax effect, most of them operating in one-dimensional space, i.e. after azimuthal integration of the diffraction frame. Other approaches consist in ensuring photons arrive in the sensitive layer orthogonally, either by using cylindrical, spherical or multiple modules aligned along a circle. The third section presents the physics of volumetric photon absorption, which is used to simulate the effect. In the fourth section, we introduce the averaged parallax effect which compensates for the displacement of the rings. In the fifth section, the usage of this correction is discussed together with its limitations.

2. State of the art

The parallax effect has been known for a long time (Hammersley *et al.*, 1994), since this effect is proportional to the sensitive-layer thickness and detectors were based on ionization gas chambers (several centimetres in size) before the advent of CCD detectors coupled with phosphors or scintillators, which are much thinner.

2.1. Ensure the detector is locally orthogonal to the incident beam

The first attempts to address parallax consisted in ensuring that photons enter the sensitive layer orthogonally. This can be achieved by manufacturing detectors that are spherical (Zanevsky *et al.*, 1995) or planispherical (Brunbauer *et al.*, 2017), where the sample is located at the centre of the sphere. Another approach is to use a cylindrical detector (Tolborg *et al.*, 2017) where the sample is located at the centre of the cylinder. In this configuration, the parallax is accounted for only in one direction, not the other. Finally, this effect can also be mitigated by using several smaller detectors assembled in arrays (Pilatus 12M, Diamond), or by rotating one or several detectors on a 2θ arm (Kawaguchi *et al.*, 2024).

These methods all suffer from the same weakness: the sample needs to be precisely located at the centre of the detector geometry, and any deviation may be difficult to account for.

2.2. Correct the powder diffraction pattern a posteriori

A simple post-integration data correction has been proposed to restore the actual scattering angle by applying a polynomial correction to the expected scattering angle (Marlton *et al.*, 2019), possibly coupled with a re-binning of the data to ensure the 2θ- or q-positions are evenly spaced. This rebinning creates an extra blurring of the signal. The main drawback of the approach is that it requires the detector, assumed to be flat, to be mounted orthogonally to the incident beam, which limits the accessible q-range in comparison with tilted detectors.

2.3. Neural network based corrections

ParallaxNet (Dong *et al.*, 2024) is an autoencoder neural network designed to address the parallax effect, but it has been trained to address the variation of the sample–detector distance during a diffraction tomography experiment. The principle could be re-used to take the powder diffraction patterns from a database like SIMPOD (Rincón *et al.*, 2025) and apply the parallax effect as described in section 3.3. The ensemble of blurred and theoretical patterns could be used to train a convolutional neural network to perform the unblurring of diffraction images.

2.4. Analytical model of the average displacement

The resolution function of powder diffraction setups based on area detectors has been described by Chernyshov *et al.* (2021), where the displacement caused by the parallax effect is modelled as $\sin(\alpha)/\mu$, with α the incidence angle of the ray on the detector and μ the linear attenuation coefficient of the sensor material. The main issue with this description is that it does not explain the increase of the parallax effect observed above $30°$. This limitation is addressed in section 4.1.

3. Physics of the parallax effect

3.1. The Beer–Lambert law

The Beer–Lambert law describes the attenuation of light as it passes through matter:

$$dI = -\mu I dx \tag{2}$$

where μ is the linear attenuation coefficient and x is the path length. It is better known in its integrated form, showing an exponential attenuation of the light intensity:

$$I = I_0 e^{-\mu x} \tag{3}$$

In X-ray area detectors with thick sensitive layers, this law governs the probability of photon interaction at different depths within the detector material.

3.2. Simulation of the parallax effect using raytracing

Raytracing is an old computer vision problem which has been efficiently solved by Amanatides & Woo (1987). The implementation in pyFAI is a direct re-implementation of this seminal work, using a Cython-compiled extension for performance (Behnel *et al.*, 2011).

3.2.1. Implementation of raytracing in pyFAI The software library pyFAI contains a module which performs raytracing, casting several rays per pixel of the detector and measuring, for each ray, the length of the trajectory in each volumetric pixel (voxel).

Subsequently, the Beer–Lambert law is applied to decide how much energy is deposited in each voxel.

All rays cast in a pixel are averaged together and the result is stored in a sparse matrix, Φ, where one pixel in the input image contributes to several pixels in the output image. Here is an example of the usage of this module, which relies on NumPy (Harris *et al.*, 2020) and on the sparse-matrix algebra of SciPy (Virtanen *et al.*, 2020):

```
import numpy
from scipy.sparse import csc_array
import pyFAI
from pyFAI.ext.parallax_raytracing import Raytracing

ai = pyFAI.load("geometry.poni")
size = numpy.prod(ai.detector.shape)
rt = pyFAI.ext.parallax_raytracing.Raytracing(ai)

sparse = csc_array(rt.calc_csr(32),  # throw 32x32=1024 rays per pixel
                   shape=(size, size))
```

In this example, the *geometry.poni* file contains the detector position and description, including the sensor material, thickness and the wavelength used. It is the typical PONI-file produced by the *pyFAI-calib2* application, where the sensor materials and thicknesses of all known detectors have been tabulated (~300 detectors).

3.3. Blurring of a synthetic powder diffraction frame

The `sparse` matrix obtained in section 3.2.1 describes the contribution of every input pixel to the output pixels. Applying the parallax effect to a synthetic image, i.e. blurring it, therefore reduces to a simple multiplication of this matrix Φ with the raw image r, considered as a vector:

$$p = \Phi \cdot r \tag{4}$$

which can be implemented in Python (van Rossum, 1989) as:

```
parallax_image = sparse.dot(raw_image.ravel()).reshape(raw_image.shape)
```

Here `raw_image` is a synthetic powder diffraction frame (r): the `ravel()` method flattens the 2D image into a vector, which is multiplied by the sparse matrix Φ, and the result is reshaped back to the original image shape: `parallax_image`.

3.4. Correction of the parallax effect

The initial signal before the parallax effect could theoretically be retrieved by inverting the matrix Φ. Despite this matrix being square, it is not invertible, making this inversion an ill-posed problem (Hadamard, 1902), which can be addressed using inverse methods that try to solve the following equation:

$$\hat{r} = \arg\min_r \|\Phi \cdot r - p\|^2 + \lambda R(r) \tag{5}$$

where the first term is the data term and the second the regularization term, with λ the regularization parameter and R the penalty function. This inversion of the parallax effect is highly sensitive to the noise present in the measured image, which is inevitable, notably Poisson noise.

Several pseudo-inversion tools, such as Tikhonov regularization (Tikhonov, 1963), iterative least-squares solvers like LSMR (Fong & Saunders, 2010), or the maximum-likelihood expectation-maximization (MLEM) solver for Poissonian data (Shepp & Vardi, 1982), have been tested. These investigations are available as part of the pyFAI documentation (Kieffer, 2025*a*).

Inverse methods are an active field of research in mathematics, and the methods tested (LSMR and MLEM) all turned out to be slow (typically tens of seconds of processing per 4-Mpixel frame) and to introduce artefacts in the corrected images. Fewer artefacts were observed when using the MLEM algorithm.

The use of inverse methods is impractical due to their computational requirements, and so is the approach proposed in Chernyshov *et al.* (2021), which requires a precise description of the sample and beam shape, as implemented in pyFAI's documentation (Kieffer, 2025*b*). We have thus searched for another approach that would be applicable to the parallax issue.

4. Average parallax effect

4.1. Calculation of the displacement

The average displacement of the detected signal caused by the parallax effect can be calculated analytically for a flat sensor of thickness t (figure 3). Let α be the incidence angle of the X-ray on the sensor, measured from the normal to its surface, z the depth inside the sensor below the point of incidence (POI) and x the lateral position on the sensor surface, measured along the projection of the ray.

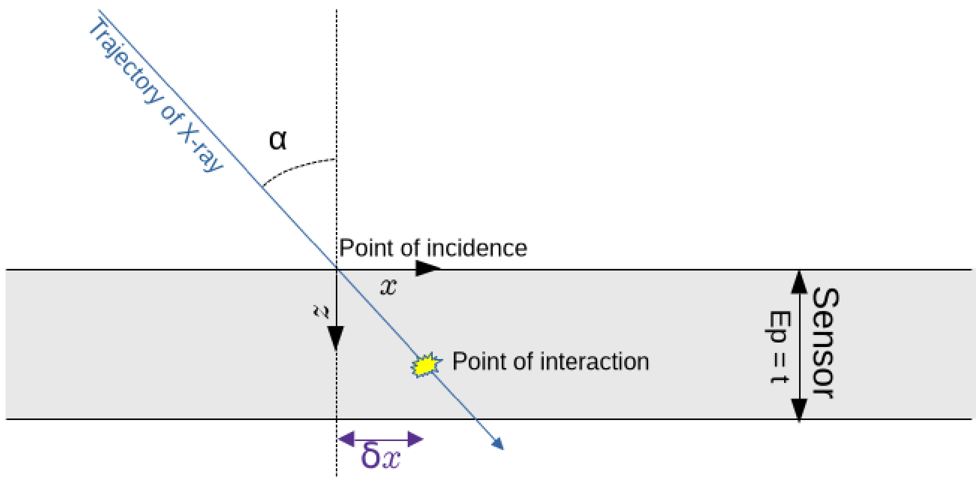

Fig. 3. Absorption in the volume of the sensor: axis definition.

The average displacement $\langle x \rangle$ of the deposited energy is the expectation value of the observable $x(z)$:

$$\langle x \rangle = \int_0^t x(z) \cdot P(z)\, dz \tag{6}$$

where $P(z)$ is the probability density for the photon to interact at depth z. Geometrically, one has:

$$x(z) = z \tan(\alpha) \tag{7}$$

and $P(z)$ follows the Beer–Lambert law (equation 3), the path travelled in the sensor to reach the depth z being $z/\cos(\alpha)$:

$$P(z) = A\, e^{\frac{-\mu z}{\cos(\alpha)}} = A\, e^{-\mu' z} \tag{8}$$

with $\mu' = \mu/\cos(\alpha)$ the apparent attenuation coefficient along the depth and A a normalization constant. Since a signal has been recorded, the photon is known to have interacted somewhere within the sensor thickness, so P is normalized over $[0, t]$:

$$1 = \int_0^t P(z)\, dz \tag{9}$$

which leads to:

$$P(z) = \mu' \cdot \frac{e^{-\mu' z}}{1 - e^{-\mu' t}} \tag{10}$$

After integration of equation 6, the average displacement becomes:

$$\langle x \rangle = \frac{\sin(\alpha)}{\mu} \cdot \frac{1 - (1 + \mu' t)\, e^{-\mu' t}}{1 - e^{-\mu' t}} \tag{11}$$

This average displacement depends only on quantities available in the geometry description of the experimental setup: the incidence angle of the ray, the linear attenuation coefficient of the sensor material at the given photon energy and the sensor thickness.

The numerical value of this displacement is compared in figure 4 with the simulation obtained by convolving a Gaussian-shaped beam profile with a truncated exponential decay. The curve from equation 11 shows a remarkable similarity to the numerically integrated displacement from Kieffer (2025*b*).

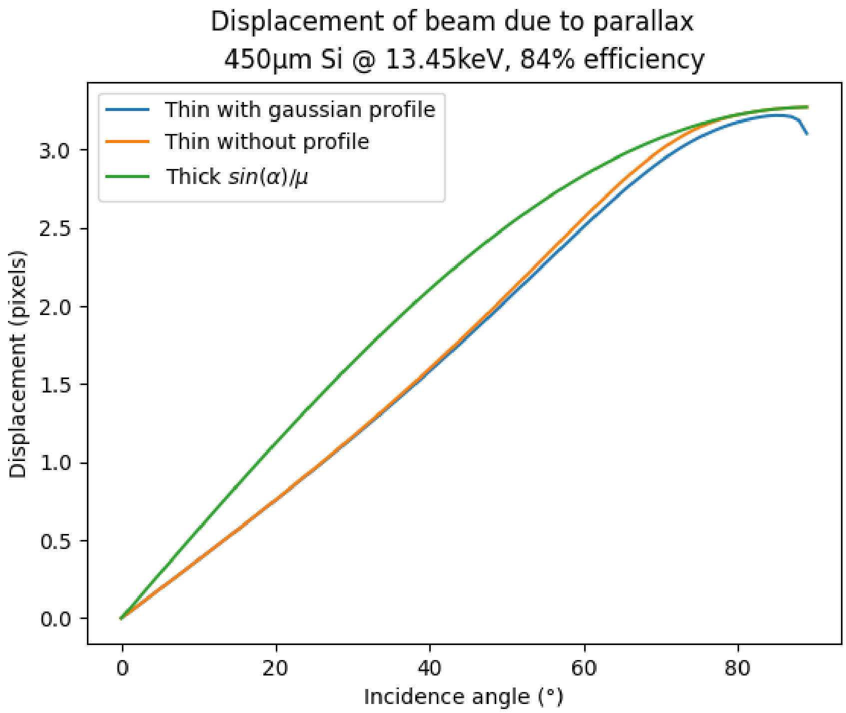

Fig. 4. Average distance between the point of incidence and the point of interaction as a function of the incidence angle, in pixels (75 μm) for a 450 μm silicon sensor and an X-ray beam at 13.45 keV. The orange curve corresponds to equation 11 (this work), the green curve to Chernyshov *et al.* (2021) and the blue curve to the numerical convolution (Kieffer, 2025*b*).

It is worth noting that if the sensor is assumed to be infinitely thick ($t \to \infty$), the same integral leads to the simpler expression (12), which was already given by Chernyshov *et al.* (2021), albeit without derivation (see section 2.4).

$$\langle x \rangle = \frac{\sin(\alpha)}{\mu} \tag{12}$$

4.2. Correction of the displacement

The correction consists in shifting the position of every pixel by the average displacement $\langle x \rangle$ given by equation 11, in the direction opposite to the propagation of the ray. In pyFAI, this correction is obtained by linear interpolation from the scattering angles calculated after correction.

4.3. Application to synthetic data

Figure 5 illustrates the correction on a synthetic powder diffraction frame, azimuthally integrated (left panel) and zoomed on a single peak (right panel). The blue curve corresponds to a pattern generated from the crystallographic information of the calibrant (SRM640 Silicon), with a very narrow width, from which a diffraction image is generated. The orange curve corresponds to the azimuthal integration of this image: the peak remains well centred on the theoretical position. The green curve corresponds to the application of the parallax effect (section 3.3), which displaces the peak towards higher scattering angles. Finally, the red curve corresponds to the correction described in section 4.1: the maximum of the peak is back at the theoretical position, but the broadening induced by the parallax remains present.

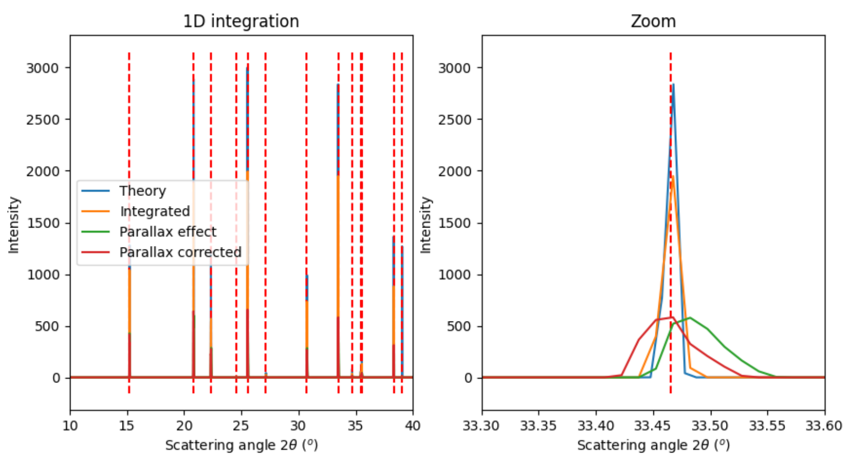

Fig. 5. Effect of the parallax correction on a synthetic powder diffraction frame of the silicon calibrant with an Eiger2 4M detector at 25 keV: azimuthal integration (left) and zoom on a single peak at large incidence angle (right). The dashed vertical lines indicate the theoretical peak positions.

4.4. *Application to experimental data* (CeO_2 *on CdTe & Si detectors)*

5. Discussion

5.1. *Usage in pyFAI*

The sensor material and thickness are part of the detector description in pyFAI, and the correction can be activated directly from the *pyFAI-calib2* graphical user interface, as shown in figure 6.

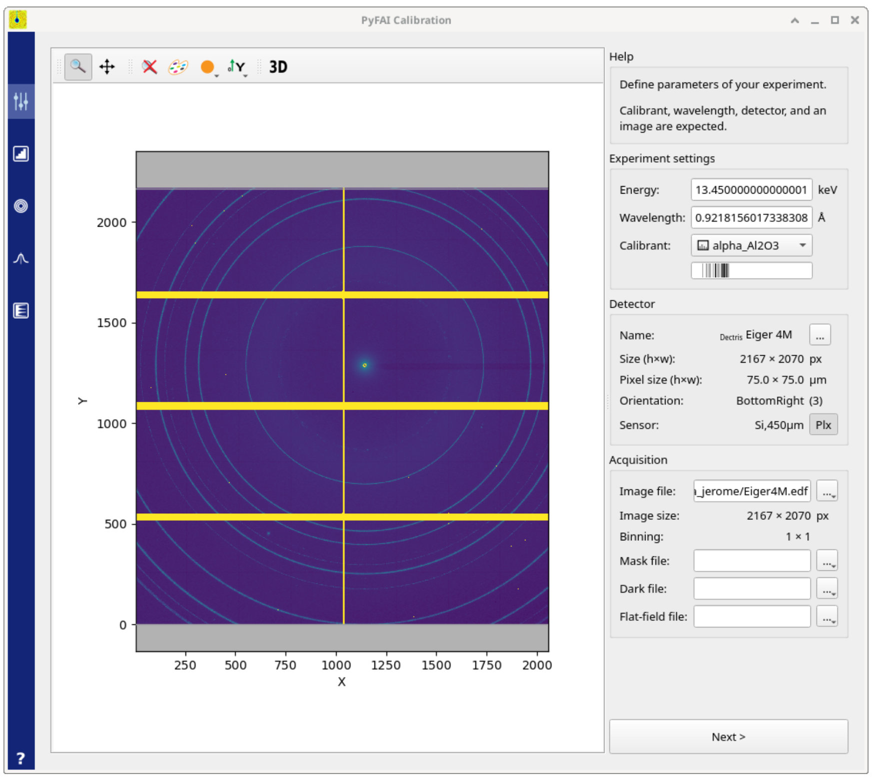

Fig. 6. Experiment-settings tab of the *pyFAI-calib2* graphical user interface. The sensor material and thickness (here Si, 450 μm) are displayed in the detector description, and the *Plx* button activates the parallax correction described in section 4.1.

5.2. *Limitations of the average model*

Although the peak is now located at the proper position, the peak broadening caused by the parallax effect cannot be corrected without the use of an inverse method (see section 3.4). The detector thickness is only one contribution to peak broadening: sample height and width, pixel size, beam divergence and wavelength dispersion are also important and are not taken into account.

5.3. Application to powder diffraction and PDF measurement

This new method of taking the parallax into account makes it possible to tilt the detector in order to increase the accessible diffraction angles without further distorting the peak shapes. Indeed, until now the parallax effect has mostly been taken into account within (Rietveld) refinements, which require all diffraction angles to share the same incidence angle on the detector, hence a detector mounted orthogonally to the incident beam. This constraint greatly limits the usability for PDF analysis. With this correction applied at the azimuthal-integration stage, the peaks are still broadened by the parallax, but less than if the different pixels contributing to the same peak had different incidence angles.

6. Conclusion

The parallax effect, i.e. the volumetric absorption of photons within the thickness of the sensor, displaces and broadens the diffraction peaks recorded with 2D area detectors, degrading the quality of geometry calibrations and of the extracted diffraction data at scattering angles beyond 30°.

Two complementary approaches to cope with this effect, both implemented in the pyFAI library, have been presented. On the one hand, ray tracing through the sensor volume, driven by the Beer–Lambert law, models the effect as a sparse matrix which allows the blurring of synthetic diffraction images. The pseudo-inversion of this matrix can in principle restore the unperturbed image, but the problem is mathematically ill-posed: the inverse methods tested require tens of seconds of computation per image and introduce artefacts in the corrected images. On the other hand, an analytical expression of the average displacement was derived. It depends only on the incidence angle of the ray, the linear attenuation coefficient of the sensor material and the sensor thickness, all of which are available in the geometry description of the experimental setup. Applied at the azimuthal-integration stage as a correction of the pixel positions, it restores the peak maxima to their theoretical positions regardless of the detector position and orientation, at a negligible computational cost; only the parallax-induced broadening of the peaks remains.

This correction, integrated in the geometry engine of pyFAI and directly activatable from the *pyFAI-calib2* graphical user interface, relaxes the constraint of mounting the detector orthogonally to the incident beam. It thereby opens the way to tilted-detector geometries which extend the accessible q-range, an asset for powder diffraction and especially for pair distribution function analysis, where thick, high-efficiency sensors and high scattering angles are both required.

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Synopsis

Numerical simulation and correction of the 2θ values due to the parallax effect, i.e. volumetric absorption of photons in the thickness of the sensor layer.